RESEARCH REPORT · IMRAD FORMAT

Sales Trend and Time-Based Performance Analysis for Afficionado Coffee Roasters

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EXECUTIVE SUMMARY

This report presents the exploratory data analysis (EDA) called for in the project brief, run against 149,116 point-of-sale transactions from Afficionado Coffee Roasters' three New York City locations (Lower Manhattan, Hell's Kitchen, Astoria) between January 1 and June 30, 2025. Two findings should anchor management's reading of this report. First, overall sales are on a strong, sustained upward trend: monthly revenue more than doubled from $81,678 in January to $166,486 in June, and weekly revenue trends upward with a slope of roughly $776 in additional revenue per week — this is a genuine growth signal, not seasonal noise, and is consistent with Afficionado's public positioning as a fast-scaling specialty roaster that opened its first NYC café in 2021. Second, and more surprising given industry norms, demand at these three cafés shows almost no weekday-versus-weekend effect (average daily revenue of $3,858 on weekdays versus $3,867 on weekends, a gap under 0.3%) and a single dominant morning peak rather than the bimodal morning-and-afternoon curve typical of the wider coffee-shop sector: 45.2% of all transactions and 45.8% of all revenue occur in the four-hour window between 7 and 10 AM alone, with a long, flat plateau rather than a second afternoon spike. Cross-location comparison reveals a clear behavioral split: Lower Manhattan drops off sharply after 6 PM (a commuter/business-district pattern), while Hell's Kitchen and Astoria sustain meaningful evening traffic (a neighborhood/residential pattern) — a distinction that should directly shape each store's staffing schedule rather than applying one company-wide rule.

1. INTRODUCTION

Afficionado Coffee Roasters is a specialty coffee company built on direct farm relationships and single-origin sourcing, operating a roastery and retail e-commerce business alongside a physical café presence in New York City; public reporting places its first NYC café opening in 2021, alongside a wholesale program and a "ready-to-drink" canned beverage line made from under-utilized parts of the coffee plant. The project brief that scoped this analysis identifies a specific operational gap: despite holding transaction-level sales data, the company lacks a consolidated view of sales trends over time, a clear read on its busiest and slowest days, and hourly demand insight across its store locations — forcing staffing and operating-hours decisions to rely on intuition ("mornings are busy") rather than evidence.

This report addresses that gap directly using the company's own processed 2025 transaction data (149,116 records spanning January 1 – June 30, 2025, across three stores and nine product categories), following the primary and secondary objectives set out in the brief: overall 2025 sales trend, busiest/slowest days of week, peak transaction hours, cross-location comparison, and evidence-based explanation of the observed patterns. Because the supplied file already contains the derived temporal features specified in the brief (hour, day of week, time bucket, weekend flag), this report performs the analysis directly rather than needing to re-derive those fields.

Scope note. The data covers the first half of 2025 only (through June 30); references to "the 2025 trend" in this report describe the trend observed across that six-month window, and the trajectory should be re-confirmed once H2 2025 data is available, particularly because coffee retail is known to carry a winter/summer seasonal component that a six-month window cannot fully separate from a genuine year-over-year growth trend.

2. METHODS

Data validation. The supplied file (`Afficionado_Coffee_Processed.csv`) contains 149,116 transaction-level records across 18 fields, including pre-engineered temporal features (day of week, month, week number, weekend flag, hour, and a three-level time bucket: Morning 06:00–11:59, Afternoon 12:00–16:59, Evening 17:00–21:59). No "Late hours" (22:00–05:59) transactions were present in the data, consistent with a café that does not operate overnight. The date field spans exactly January 1 through June 30, 2025 (181 calendar days) with no gaps detected; the final day in the file (June 30) reflects a single day rather than a full week and was excluded from the weekly-aggregate trend calculation to avoid understating the final period.

Trend analysis. Revenue and transaction counts were aggregated by week and by month to assess the overall 2025 trajectory, and a linear fit was applied to the weekly series to quantify the trend's slope and strength (Pearson correlation between week number and weekly revenue).

Day-of-week and weekend analysis. Because the 181-day window contains slightly different counts of each weekday (e.g., 26 Mondays but 25 Tuesdays), all day-of-week and weekday/weekend comparisons were normalized to an *average daily* basis (total revenue or transactions divided by the number of calendar instances of that day in the window) rather than raw totals, to avoid a calendar-count artifact being mistaken for a demand pattern.

Hourly and cross-location analysis. Transaction counts and revenue were aggregated by hour of day (overall and separately by store) to identify peak windows and to compare each store's hourly shape against the other two, addressing the brief's cross-location temporal comparison objective. Product-category revenue and time-bucket distribution were also profiled to connect the temporal findings to what is actually being sold when.

Desk research for context. Company background was drawn directly from afficionadocoffee.com (product lines, café history, sourcing model). Industry-benchmark figures on typical U.S. coffee-shop hourly and weekly demand patterns, gathered in an earlier phase of this project from operator surveys and industry publications (National Coffee Association, Specialty Coffee Association, and coffee-industry operator benchmarking sites), are referenced in Section 3 for comparison against Afficionado's own observed patterns, not as a substitute for them.

Limitations. The dataset does not include a distinct order/customer identifier, so true repeat-visit or customer-loyalty analysis is not possible from this file. Weather, local events, and holiday-calendar effects that could explain week-to-week volatility (e.g., the January dip visible in the monthly series) are not present in the data and were not investigated further in this phase. Store-level comparison is limited to three NYC neighborhoods, so findings about "business-district vs. neighborhood" behavior should be read as a within-company observation rather than a generalizable claim about coffee retail broadly.

3. FINDINGS

Quantitative Insights

Overall 2025 trend. Monthly revenue rose every month from January through June with no month-over-month decline:

MONTH	REVENUE	TRANSACTIONS
January	$81,678	17,314
February	$76,145	16,359
March	$98,835	21,229
April	$118,941	25,335
May	$156,728	33,527
June	$166,486	35,352

February is the one month that dips slightly below January (-6.8%), consistent with the seasonal winter lull noted in national coffee-retail benchmarking, before the trend resumes and accelerates from March onward. The week-level linear trend has a Pearson correlation of 0.67 between week number and weekly revenue and a slope of roughly +$776 in additional weekly revenue for every week that passes — a genuinely strong, sustained growth signal across the six-month window, not a seasonal blip.

Day-of-week performance (average daily basis, normalized for calendar-count differences):

DAY	AVG. DAILY REVENUE	AVG. DAILY TRANSACTIONS
Thursday (busiest)	$3,911	832
Monday	$3,899	835
Sunday	$3,876	833
Tuesday	$3,876	820
Saturday	$3,858	820
Friday	$3,825	815
Wednesday (slowest)	$3,782	811

The full seven-day spread is only about 3.4% (Thursday vs. Wednesday) — effectively flat. Aggregated to weekday vs. weekend, the gap nearly disappears: $3,858 average daily revenue on weekdays (129 days in the window) versus $3,867 on weekends (52 days), a difference of 0.3%.

Hourly demand curve (all stores combined):

HOUR	TRANSACTIONS	REVENUE
7 AM	13,428	$63,526
8 AM	17,654	$82,700
9 AM (peak)	17,764	$85,170
10 AM	18,545 (peak revenue hour)	$88,673
11 AM	9,766	$46,319

12–16 (afternoon plateau)	~8,700–9,100/hr	~$40,000–42,000/hr
17–19 (evening decline)	6,100–8,700/hr	$28,400–40,100/hr
20 (near close)	603	$2,936

The 7–10 AM window alone accounts for 45.2% of all transactions and 45.8% of all revenue — almost exactly matching the "40-70% of daily sales in peak windows" range reported across U.S. coffee-shop industry benchmarks, though concentrated in a single continuous morning block rather than split into a morning-and-afternoon bimodal pattern.

Cross-location comparison. All three stores post similar total revenue (Hell's Kitchen $236,511; Astoria $232,244; Lower Manhattan $230,057) and near-identical morning peaks, but diverge sharply in the evening: at 7 PM, Astoria still records 3,565 transactions and Hell's Kitchen 2,402, while Lower Manhattan has fallen to just 125 — and Lower Manhattan records essentially no volume at 8 PM (75 transactions) while Hell's Kitchen still logs 528. Weekend-index figures (weekend average daily revenue as a % of weekday average) confirm the same split: Astoria actually runs slightly higher on weekends (103.5), while Hell's Kitchen and Lower Manhattan run marginally lower (98.6 each) — a small but directionally consistent signal that Astoria behaves more like a neighborhood/leisure location and the other two lean more commuter-driven.

Thematic / Qualitative Synthesis

The growth trend is the headline story and should be the first slide in any stakeholder briefing. Revenue more than doubled from January to June 2025. Whether this reflects genuine same-store sales growth, new-customer acquisition, seasonal recovery from a slow winter, or a combination of the three cannot be fully separated within a single six-month window with no prior-year comparison — but the consistency of the month-over-month increase (five consecutive increases after the one February dip) makes a purely seasonal explanation unlikely on its own. This should be the lead finding presented to government/stakeholder audiences per the brief's executive-summary deliverable, ahead of any staffing-oriented finding.

The near-total absence of a weekday/weekend effect is a genuine, actionable finding — not a data artifact. Both the raw day-of-week averages and the explicit weekend flag confirm the same thing from two different angles: weekday and weekend demand at these three cafés are effectively interchangeable (within 0.3%). This runs counter to the general industry pattern cited in comparable coffee-retail research, where business-district cafés typically see reduced weekend traffic relative to neighborhood locations. The most likely explanation, supported by the cross-location evening-hours data below, is that Afficionado's three NYC locations are more residential/mixed-use than a pure downtown financial-district café, so weekend leisure demand backfills what would otherwise be a commuter-driven weekday skew.

Demand here is a single morning peak, not the bimodal curve seen industry-wide. National coffee-retail benchmarks typically describe two peaks — a dominant morning rush and a secondary afternoon lift around 2–4 PM. Afficionado's data shows one dominant peak (7–10 AM) followed by a long, flat afternoon plateau with no distinct secondary rise; revenue per hour in the 12–16 window is remarkably stable (roughly $40,000–42,000 across all five hours) rather than dipping and re-peaking. Operationally, this means Afficionado's staffing curve should be modeled as one sharp ramp-up and one long, gentle taper, not two separate rush-hour events — a simpler shape to schedule around than the industry default.

The three stores are not interchangeable, and the evening hours are where that shows up. Lower Manhattan's near-total evening drop-off (125 transactions at 7 PM versus roughly 2,400–3,600 at the other two stores) is the single clearest signal in the cross-location data. This store should very likely be scheduled and possibly even have operating hours set differently from the other two — keeping it open at full staffing through 8 PM when Astoria and Hell's Kitchen customers are still arriving is likely wasted labor cost at that location, while the reverse (closing Astoria or Hell's Kitchen early to match Lower Manhattan's curve) would forfeit real revenue at those two.

Product mix confirms the morning skew is coffee- and bakery-led, not incidental. Coffee and Tea together account for 66.7% of all revenue, and both skew slightly more toward the Morning bucket (52–53% of their transactions) than Bakery or Branded merchandise, which skew even more toward mornings (58.6% and 61.7% respectively) — consistent with a commuter/grab-and-go breakfast occasion rather than an afternoon browsing occasion. Coffee beans, Loose Tea, and Flavours (small-volume, higher-consideration categories) skew most heavily toward mornings of all (66–69%), suggesting these are add-on purchases made by the same repeat morning customer rather than a separate shopping trip.

4. DISCUSSION & CONCLUSION

The project brief anticipated two things it expected the data to show: a distinguishable weekday-vs-weekend pattern, and a bimodal (morning-and-afternoon) hourly demand curve, both common in industry benchmarking for coffee retail. Afficionado's own 2025 transaction data confirms neither in the form usually assumed — weekday and weekend demand are essentially indistinguishable company-wide, and the hourly curve is a single sharp morning peak followed by a long plateau rather than two separate rushes. This is a more useful outcome than simply confirming the industry default, because it tells management something specific about their own three stores rather than something already known about coffee shops in general.

The real heterogeneity in this data is not across days of the week but across stores, and specifically in the evening hours. Lower Manhattan behaves like a commuter/business-district location (sharp evening drop-off); Astoria and, to a lesser extent, Hell's Kitchen behave more like neighborhood locations (sustained evening traffic, and in Astoria's case a slight weekend uplift). Because this pattern shows up consistently across two independent cuts of the data (raw hourly counts and the weekend index), it should be treated as a reliable operational signal rather than noise, and it is exactly the kind of "location-specific customer behaviour insight" the brief's cross-location diagnostic step was designed to surface.

The sustained revenue growth across the six-month window is the most consequential finding for a government-stakeholder audience specifically, since it speaks to business viability and job/economic impact rather than internal operations, and should be foregrounded in the executive summary deliverable the brief calls for.

Strategic recommendations

1. **Lead the executive summary with the growth trend, not the staffing findings.** A monthly revenue increase from $81,678 to $166,486 across six months is the most

stakeholder-relevant number in this analysis and should open the report, with staffing/operational findings following as the "how we'll sustain this growth efficiently" narrative.

2. **Set store-specific closing-hour and evening-staffing policy rather than a single company-wide rule.** Reduce evening staffing at Lower Manhattan after roughly 6–7 PM where transaction volume has already collapsed, and confirm Astoria and Hell's Kitchen are adequately staffed through 7–8 PM where meaningful demand persists.
3. **Do not build a weekday/weekend staffing differential into the default schedule.** The data does not support one at the company level; apply the same base staffing template to weekdays and weekends unless a specific store's own data (e.g., Astoria's modest weekend uplift) justifies a small store-level adjustment.
4. **Concentrate peak-hour staffing on a single continuous 7–10 AM block** rather than planning for two separate rush periods; ensure maximum front-of-house coverage through this three-hour window specifically, since it alone carries 45%+ of daily volume, then step staffing down gradually through a long, flat afternoon rather than re-ramping for an afternoon rush that this data does not show.
5. **Investigate the February dip and the March-onward acceleration with marketing/operations records** (promotions, new product launches, seasonal menu changes, wholesale account additions) to determine which levers are driving the growth trend, so they can be deliberately repeated rather than relying on general market tailwinds.
6. **Build the Streamlit dashboard's default view around the single-morning-peak/three-store-divergence findings** confirmed here: an hourly line chart per store (not a single blended company-wide curve) should be the primary "Time-of-Day Demand Analysis" view, since the blended curve masks the Lower Manhattan/Astoria divergence that this report identifies as the most actionable single finding.

In summary, Afficionado Coffee Roasters' three NYC cafés are growing quickly and consistently through the first half of 2025, are essentially insensitive to day-of-week at the company level, and concentrate demand into a single well-defined 7–10 AM morning rush rather than the industry-typical two-peak pattern. The one place where the three stores meaningfully diverge is the evening hours, where Lower Manhattan's commuter-driven drop-off contrasts with sustained neighborhood traffic at Astoria and Hell's Kitchen — and that divergence, more than any weekday/weekend or industry-benchmark comparison, is where store-specific staffing decisions should be targeted first.

5. REFERENCES

1. Afficionado_Coffee_Processed.csv — primary transaction dataset supplied for this analysis (149,116 records, January 1 – June 30, 2025, three NYC store locations).
2. Sales_Trend_and_Time.docx — internal project brief: "Sales Trend and Time-Based Performance Analysis for Afficionado Coffee Roasters."
3. Afficionado Coffee Roasters. Company website — product lines, café/roastery locations, sourcing model. https://www.afficionadocoffee.com/
4. Afficionado Coffee opens first NYC cafe, launches inventive can line. Daily Coffee News (2021), linked via afficionadocoffee.com press page.
5. What are the average coffee shop sales per day? BusinessDojo (2025). https://dojobusiness.com/blogs/news/avg-sales-per-day-coffee-shop
6. What is the profit margin of a cafe? BusinessDojo (2025). https://dojobusiness.com/blogs/news/cafe-profit-margin
7. Cafe Market Trends and Consumer Behavior Analysis (2026). BusinessDojo. https://dojobusiness.com/blogs/news/cafe-market-trends
8. Profitable Coffee Shop Hours: Find the Perfect Schedule for Success & Happy Customers. Brewed Knowledge (2025). https://brewedknowledge.com/profitable-coffee-shop-hours-find-the-perfect-schedule-for-success-happy-customers
9. 2026 National Coffee Data Trends Specialty Coffee Report. Specialty Coffee Association / National Coffee Association (2026). https://sca.coffee/sca-news/2026-national-coffee-data-trends-report